

XIAONAN GUO

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Personal Website ◇ Google Scholar

EDUCATION

Zhejiang University
B.S. in Clinical Medicine
M.D. in Psychiatry

Hangzhou, Zhejiang
Aug 2019 – Jun 2024
Sept 2024 – Jun 2027 (expected)

RESEARCH INTERESTS

- **Psychiatric Genomics:** Dissecting the genetic basis of psychiatric disorders via multi-trait GWAS to advance the clinical translation of polygenic risk scores.
- **Single-cell & Spatial Multi-omics:** Integrating snRNA-seq, Stereo-seq, and scATAC-seq to map cell-type-specific dysregulation in postmortem bipolar disorder brain.
- **Computational Psychiatry** (long-term goal): Building computational models of mood disorders to bridge molecular findings with clinical phenotypes.

RESEARCH EXPERIENCE

Research Assistant, Cui Lab

Aug 2020 – Jul 2023

Supervisor: Prof. Yihui Cui, Zhejiang University

- Participated in the project "*Hypothalamus-habenula potentiation encodes chronic stress experience and drives depression onset*" (published as *Neuron* cover); contributed to behavioral experiments under neural manipulation.
- Conducted omics analysis in a project on synaptic homeostasis regulating depression (*Nature*).

Research Assistant, Hu Lab

Sept 2020 – Present

Supervisor: Prof. Shaohua Hu, Zhejiang University

- Contributed to explorations in shared genetic architecture and bidirectional clinical risks within the psychometabolic nexus (published in *eBioMedicine*).
- *Ongoing:* Profiling cell-type-specific transcriptomic and epigenomic dysregulation in postmortem bipolar disorder brain using single-cell (snRNA-seq), spatial transcriptomics (Stereo-seq), and scATAC-seq.
- *Ongoing:* Developing polygenic risk scores to differentiate major depressive disorder from bipolar disorder.

Other Research Engagements

- Psychological research on working memory and key information effects under PI Hui Chen (Mar 2022–Apr 2022).

SELECTED PUBLICATIONS

1. Guo, X., et al. (2024). *Shared genetic architecture and bidirectional clinical risks within the psychometabolic nexus*. **EBioMedicine**. (First author)
2. Feng, Y.[†], Guo, X.[†], et al. (2026). *Cross-ancestry Pleiotropic Analysis of Imaging-derived Phenotypes Enhances Risk Stratification of Depression*. **Molecular Psychiatry**. (Co-first author; in press)
3. Feng, Y.[†], Guo, X.[†], Huang, P.[†], et al. (2026). *Cerebrospinal Fluid Genetics Enhance Risk Stratification in Bipolar Disorder*. **MedComm**, 7(3), e70629. (Co-first author)
4. Guo, X., et al. (2024). *Constructing a Theoretical Model to Bridge Neural Transition with a State Switch in Bipolar Disorder*. **Neuroscience Bulletin**. (First author)
5. Xi, C.[†], Lu, B.[†], Guo, X.[†], et al. (2024). *Characteristics of Brain Network Connectome and Connectome-Based Efficacy Predictive Model in Bipolar Depression*. **Molecular Psychiatry**. (Co-first author)
6. Wang, D.[†], Guo, X.[†], et al. (2024). *Efficacy and Safety of Transcranial Direct Current Stimulation as an Add-On Trial Treatment for Acute Bipolar Depression Patients With Suicidal Ideation*. **CNS Neuroscience & Therapeutics**. (Co-first author)
7. Guo, X., et al. (2024). *Serum signature of antibodies to *Toxoplasma gondii*, rubella virus, and cytomegalovirus in females with bipolar disorder: A cross-sectional study*. **Journal of Affective Disorders**. (First author)
8. Guo, X., et al. (2024). *Is muscarinic receptor agonist effective and tolerant for schizophrenia?*. **BMC Psychiatry**. (First author)

9. **Guo, X.**, et al. (2024). *Impact of second-generation antipsychotics monotherapy or combined therapy in cytokine, lymphocyte subtype, and thyroid antibodies for schizophrenia: a retrospective study.* **BMC Psychiatry**. (First author)

10. Kong, L.[†], **Guo, X.**[†], et al. (2024). *Pushing the Frontiers: Optogenetics for Illuminating the Neural Pathophysiology of Bipolar Disorder.* **International Journal of Biological Sciences**. (Co-first author)

PROJECTS

MindQuest	Creator, 2026
Built a free, open-source, bilingual (Chinese/English) online psychological self-assessment platform for public mental health awareness.	

ACADEMIC ACTIVITIES

Selected Member , CLS–CIBR–IDG Summer Class	Aug 2021, Beijing, China
Served as team leader; organized and presented group research.	
Co-Founder , <i>Univeron</i>	Jan 2020 – Present, Hangzhou, China
- Established a nonprofit academic organization focused on neuroscience discussions. - Organized 30+ seminars and invited international professors; built a 200+ member community.	
Teaching Assistant , Neurophysiology Course	Sept 2020 – 2022
Assisted Prof. Yihui Cui in neurophysiology experiments and student instruction.	
Clinical Rotation , First Affiliated Hospital of Zhejiang University	Aug 2022 – Present
- Holds a physician license; - Group leader; rotated through psychiatry, neurology, and neurosurgery departments; - Completed Professional Training in Psychological Counseling (Chinese Academy of Sciences), Cognitive Behavioral Therapy (CBT), Interpersonal Psychotherapy (IPT), and Mindfulness-Based Cognitive Therapy (MBCT).	

AWARDS & HONORS

1. National Scholarship (Oct 2025)	
2. Second Prize, Zhejiang University Lin Baixin Science & Technology Award (2026)	
3. Nomination for Xinglin Star, Zhejiang University School of Medicine (May 2025)	
4. Outstanding Graduate, Zhejiang University (Jun 2024)	
5. Gold Medal, 14th Zhejiang Province College Student Entrepreneurship Competition (May 2024)	
6. Gold Medal, International College Students' Innovation and Entrepreneurship Competition (Dec 2023)	

TECHNICAL SKILLS

Experimental Skills	Molecular biology, optogenetics, behavioral neuroscience
Statistical Genetics	Post-GWAS analysis (multi-trait analysis, polygenic risk scoring)
Single-cell & Spatial Omics	snRNA-seq and spatial transcriptomics analysis
Neuroimaging Analysis	Structural and functional MRI analysis (SPM, FreeSurfer)
Software	R, Python